GHG Protocol Corporate Standard

A Corporate Accounting and Reporting Standard (Revised Edition 2015)

Key Highlights for AI System Training

Key Highlights to Bookmark:

- Table 4.1: Consolidation approaches (operational control vs equity share)
- Chapter 6: calculation methodology (combustion emissions)
- Chapter 8: base year recalculation policy
- Appendix: emission factor tables

Section 1: Consolidation Approaches (Chapter 3–4)

Template Variable: `{rag_context_ghg_consolidation}`

> ℹ️ *The consolidation approach determines WHICH emissions fall within the organizational boundary. This choice directly impacts Scope 1/2 totals reported in the sustainability report. IFRS S2 Para 29(a) and GRI 305 both reference the GHG Protocol for boundary-setting.*

Two Consolidation Approaches

Approach	Rule	When to Use
Equity Share	Account for GHG emissions proportional to the equity share (% ownership) in each operation.	Aligns with financial accounting. Best for risk/liability assessment. Comprehensive coverage of liabilities.
Operational Control	Account for 100% of GHG emissions from operations over which the company has operational control (i.e., authority to introduce operating policies).	Preferred by government regulatory programs (EU ETS). Best for performance management. Simpler data access.
Financial Control	Account for 100% of GHG emissions from operations over which the company has financial control (i.e., ability to direct financial and operating policies).	Similar to operational control but determined by financial relationship. Aligns with financial consolidation rules.

Table 4.1: Consolidation Decision Matrix

How each entity type is treated under different approaches:

Entity Type	Equity Share	Operational Control	Financial Control
Wholly-owned subsidiary	100%	100%	100%
Subsidiary (e.g., 83% owned)	83%	100% (if operator)	100%
Joint venture (50/50, joint financial control)	50%	0% (if not operator)	50%
Joint venture (operator, no financial control)	Equity %	100%	0%
Non-incorporated JV (proportionally consolidated)	Proportional %	100% (if operator)	Proportional %
Associate/affiliate (significant influence, no control)	Equity %	0%	0%
Investment (no influence)	0%	0%	0%
Franchise (no equity/control)	0%	0%	0%

Key Points for Banking Sector

- Banks typically use OPERATIONAL CONTROL approach for their own operations (branches, offices, ATMs, data centers)

- For financed emissions (Scope 3, Category 15), the EQUITY SHARE concept applies — emissions attributed proportional to financing provided (per PCAF methodology)
- Double counting can occur when two companies hold interests in the same operation and use different approaches — disclosure of chosen approach is required

Section 2: Calculation Methodology (Chapter 6)

Template Variable: `{rag_context_ghg_calculation}`

> ℹ️ *Chapter 6 defines the fundamental calculation approach. The core formula is:*
> *GHG Emissions = Activity Data × Emission Factor.*
> *This is the basis for ALL emissions calculations in the sustainability report.*

Core Calculation Formula

Component	Description
GHG Emissions (output)	Expressed in metric tonnes CO_2 equivalent (tCO2e)
Activity Data (input)	Quantitative measure of activity (e.g., litres of fuel consumed, kWh of electricity purchased, km of business travel)
Emission Factor (conversion)	Ratio relating emissions to activity (e.g., kg CO_2/litre, tCO_2/MWh). Source-specific > local grid > published default.
GWP (Global Warming Potential)	Converts non-CO_2 gases to CO_2 equivalent. CH_4 = 28, N_2O = 265, SF_6 = 23,500 (IPCC AR5 values).

Calculation Hierarchy (Most to Least Accurate)

- Direct measurement (continuous emission monitoring — CEM)
- Mass balance / stoichiometric calculation (facility/process-specific)
- Fuel-specific emission factors (fuel consumption × carbon content coefficient)
- Published default emission factors (IPCC, national inventories)

Stationary Combustion Emissions (Primary Scope 1 Source)

Formula: CO_2 Emissions = Fuel Consumption × Emission Factor

Scope	Typical Activity Data	Emission Factor Source
Scope 1 (Direct)	Purchased fuel quantities (natural gas m³, diesel litres, LPG kg)	Fuel-specific carbon content coefficients; supplier data
Scope 2 (Indirect — Energy)	Metered electricity consumption (kWh), purchased steam/heat	Supplier-specific, local grid factor, or national average
Scope 3 (Other Indirect)	Fuel use, passenger-km, tonne-km, financial data (for PCAF)	Published or third-party emission factors; PCAF factors for financed emissions

GHG Protocol Calculation Tools (Table 3)

Cross-sector tools available on ghgprotocol.org:

Tool	Main Features
Stationary Combustion	Calculates direct and indirect CO_2 from fuel combustion in stationary equipment. Provides co-generation allocation options and default emission factors.
Mobile Combustion	Calculates direct/indirect CO_2 from mobile sources (road, air, water, rail). Provides national average fuel and electricity emission factors.
HFC from Refrigeration & A/C	Calculates direct HFC emissions during manufacture, use and disposal. Three methodologies: sales-based, life cycle, emission factor-based.
Uncertainty Assessment	Quantifies statistical parameter uncertainties. Automates aggregation for basic uncertainty assessment of GHG inventory data.

Section 3: Base Year Recalculation Policy (Chapter 5)

Template Variable: `{rag_context_ghg_base_year}`

> ℹ️ *Chapter 5 "Tracking Emissions Over Time" defines WHEN and HOW to recalculate the base year. This is critical for the AI system because year-over-year comparisons (Perubahan%) in the OJK table must use consistent baselines. Incorrect base year handling = misleading trend data.*

Base Year Selection Requirements

- Companies SHALL choose and report a base year for which verifiable emissions data are available
- Can be a single year OR an average of several consecutive years (to smooth fluctuations)
- Must specify reasons for choosing that particular year
- Base year can also serve as target base year for GHG reduction targets

Mandatory Recalculation Triggers

Companies SHALL develop a base year emissions recalculation policy. The following cases SHALL trigger recalculation:

Trigger	Details
1. Structural Changes	Transfer of ownership or control of emissions-generating activities. Includes: mergers, acquisitions, divestments, outsourcing, and insourcing of emitting activities. Must recalculate even if single change is minor but cumulative effect is significant.
2. Methodology Changes	Changes in calculation methodology or improvements in accuracy of emission factors or activity data that result in significant impact on base year data.
3. Discovery of Errors	Significant errors, or a number of cumulative errors that are collectively significant.

Significance Threshold Policy

- Companies must define a "significance threshold" (qualitative and/or quantitative) that triggers recalculation
- Must disclose the threshold
- Must apply the policy CONSISTENTLY (recalculate for both increases and decreases)
- Verifier confirms adherence to threshold policy

Recalculation Methodology

For acquisitions: Add the acquired entity's emissions in the base year to the original base year total

For divestments: Subtract the divested entity's emissions in the base year from the original base year total

Organic growth/decline: Do NOT trigger base year recalculation (these reflect genuine performance changes)

> ℹ️ *AI System Rule: When generating year-over-year comparison data, the system must check if base year recalculation was triggered. If a structural change occurred, historical data in the esg_aggregated zone should reflect recalculated values (REQ-ETL-26). The LLM must use the recalculated baseline for Perubahan% calculation.*

Section 4: Emission Factor Reference (Appendix D)

Template Variable: `{rag_context_ghg_emission_factors}`

> ℹ️ *Appendix D provides emission factors by industry sector and scope. The GHG Protocol website (ghgprotocol.org) hosts regularly updated calculation tools with embedded emission factors. For Indonesian context, KLHK (Ministry of Environment) publishes national grid emission factors.*

Emission Factor Hierarchy (Most to Least Preferred)

#	Factor Type	Description & Example
1	**Source/facility-specific**	Measured or calculated for the specific facility. Highest accuracy. E.g., stack testing results.
2	**Supplier-specific**	Provided by fuel/energy supplier for the specific product. E.g., PLN grid factor for specific region.
3	**Country/regional average**	Published national or regional factors. E.g., Indonesia grid emission factor from KLHK/ESDM.
4	**International default**	IPCC or IEA published defaults. Used when country-specific data unavailable.

Key Emission Factor Categories for Banking Sector

Source	Scope	Typical Factor Source
Purchased electricity (offices, branches, data centers)	Scope 2	PLN grid emission factor (tCO_2/MWh) from KLHK
Diesel generators (backup power)	Scope 1	2.68 $kgCO_2$/litre diesel (IPCC default)
Company vehicles (fleet)	Scope 1	Mobile combustion tool (fuel-specific)

Source	Scope	Typical Factor Source
Refrigerant leakage (A/C systems)	Scope 1	HFC GWP values × leakage rate
Business travel (air, road)	Scope 3	DEFRA/ICAO emission factors by distance class
Financed emissions (loan portfolio)	Scope 3 Cat 15	PCAF database; sector-specific EFs

Six Kyoto Protocol Gases Covered

Gas	Formula	GWP (AR5, 100yr)	Typical Source
Carbon Dioxide	CO_2	1	Fossil fuel combustion
Methane	CH_4	28	Incomplete combustion, waste
Nitrous Oxide	N_2O	265	Combustion, fertilizers
HFCs	Various	12 – 14,800	Refrigeration, A/C
PFCs	Various	7,390 – 12,200	Semiconductor, aluminum
Sulphur Hexafluoride	SF_6	23,500	Electrical equipment